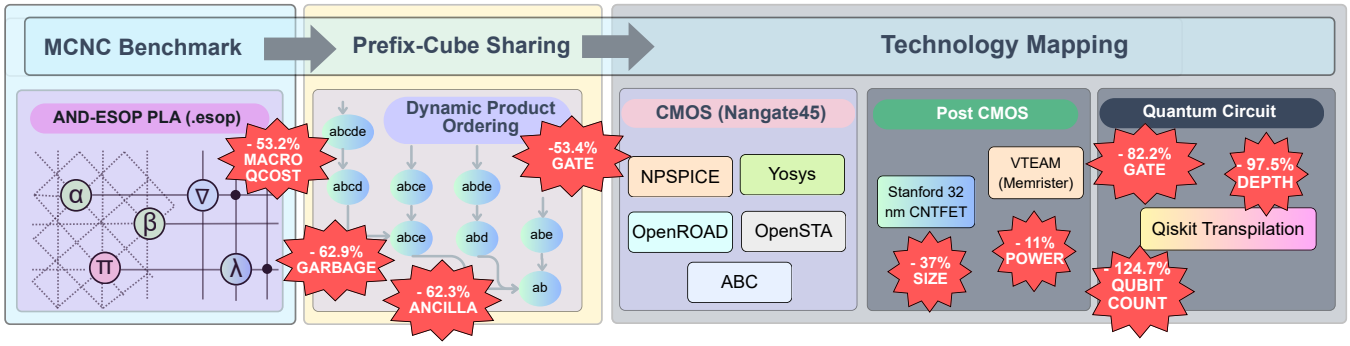
Prefix-Cube Sharing for Reversible PLA Synthesis: Gate-Minimal AND-Plane Mapping Across Quantum and Post-CMOS Technologies

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Graphical Abstract

***Abstract*—This paper presents Prefix-Cube Sharing for reversible programmable logic array (RPLA) synthesis: a gate-minimal AND-plane mapper that inserts a fixed exclusive-OR sum-of-products (ESOP) cover into a prefix-sharing tree before any device map. Under a greedy literal order, each distinct multi-literal prefix is one Universal MUX (UMG) or Universal NOR (UNG) cell, so each shared conjunction is instantiated once; the exclusive-OR plane uses the published Feynman-gate construction. The resulting netlist is interpreted across quantum and post-CMOS technologies without resynthesis: Nangate45 pass-transistor logic, Stanford carbon-nanotube field-effect transistor (CNTFET) and VTEAM cells, and a Qiskit model that reports gate count, depth, and qubit count from computational-basis readout of the Bennett embedding. On 20 representative MCNC PLA circuits, mean reversible gate count, garbage, ancilla, and macro quantum cost fall by 53.4%, 62.9%, 62.3%, and 53.2% versus the product-list RPLA. Versus a Qiskit-transpiled NOT/CNOT/Toffoli ESOP cascade, mean gate count falls from 6492.2 to 1154.0 (82.2%) and AND-plane depth from 907.5 to a bound of 22.3 (97.5%); shared prefixes occupy rails, so mean qubit count is 785.6 versus 349.6 (124.7% more qubits). Median synthesis runtime is 9.4 ms.**

***Index Terms*—Prefix-Cube Sharing, reversible PLA, AND-plane mapping, ESOP, post-CMOS, Qiskit, gate count, garbage, ancilla, circuit depth, Qubit count.**

I. Introduction

THE AND plane dominates the reversible gate count of a reversible programmable logic array (RPLA). An RPLA implements multi-output Boolean functions as Exclusive-OR sums of products (ESOPs). After a Bennett embedding the array is a computational-basis unitary; that embedding is a cost model, not a quantum speedup. Prefix Cube attains

$$C_{\wedge}(P,\sigma) = |V_{\geq 2}(T(P,\sigma))|, \tag{1}$$

one Universal MUX or Universal NOR (UMG/UNG) per distinct multi-literal prefix of a tree $T(P,\sigma)$. One netlist $\mathcal{N}$ is interpreted as Nangate45 PTL, Stanford CNTFET/VTEAM, and Qiskit [1]. Those maps do not recover switching energy [2], [3]: at room temperature $kT\ln 2$ is far below $\frac{1}{2}CV_{\mathrm{DD}}^2$, so the milliwatts remain ordinary CV^2, lowered only by fewer devices. *Prefix Cube* is this mapper, not a new cell. Table I names the resources.

A classical PLA [4]–[6] uses an AND plane and an inclusive-OR plane. Inclusive OR is not a permutation. The Feynman gate $(x, y) \mapsto (x, x \oplus y)$ is, with $\mathrm{qc} = 1$ [7]. Replacing OR by XOR gives a cover P with no fan-out except a Feynman copy. With gate set G and width w,

$$\begin{aligned} w &= n + \mathrm{AI}, \quad \mathrm{GB} = w - m, \quad \mathrm{GA} = |G|, \\ \mathcal{R}(\mathcal{N}) &= (\mathrm{GA}, \mathrm{GB}, \mathrm{AI}, \mathrm{QC}), \quad \mathrm{QC} = 5n_{\mathrm{AND}} + n_{\mathrm{XOR}}. \end{aligned} \tag{2}$$

UMG/UNG cells have $\mathrm{qc} = 5$ [8]. Bennett's embedding

$$\Phi(x,a) = \big(x,\ a \oplus f(x)\big), \qquad a \in \{0,1\}^m, \tag{3}$$

is the permutation required for the computational-basis unitary [9]–[11]

$$U_\pi|x\rangle = |\pi(x)\rangle, \qquad U_\pi^\dagger U_\pi = I_{2^w}. \tag{4}$$

The CAD task is to cut n_{AND}, hence GA, on a fixed cover P before technology mapping. Pairwise grids and product-list sorts share only locally. Prefix Cube builds the sharing tree under a greedy literal order, reuses the published Feynman XOR plane, and never builds a 2^n table. The same $\mathcal{N}$ is then scored as Nangate45 PTL, Stanford CNTFET/VTEAM, and Qiskit by gate count, depth, and qubit count [12], [13]. Section II states the comparison and the contributions; Section III develops the mapper; Section IV maps one netlist through

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TABLE I
RESOURCE NAMES USED IN THIS PAPER

Name	Symbol	Meaning
Reversible gate count	GA	$\|G\|$ of netlist $\mathcal{N}$ (UMG/UNG/Feynman and overlay)
Qiskit gate count	G	Operation count on a Qiskit circuit
Macro quantum cost	QC	$5n_{\mathrm{AND}} + n_{\mathrm{XOR}}$ on $\mathcal{N}$
Circuit depth	D	Qiskit `depth()` after the stated basis
Qubit count	Q	Qiskit `num_qubits`
Width	w	n + AI rails of $\mathcal{N}$; a *rail* becomes a *qubit* only after embedding

TABLE II
CAD COMPARISON OF PROGRAMMABLE REVERSIBLE ARRAYS

Year	Work	Input/fabric	Sharing/CAD	Metric	Map
2003	Existing [15]	2^n table	Existing NCT cascade [12]	QC	—
2006	Existing [12]	quantum circuit	NCT library	G, D, Q	Qiskit
2007	Existing [18]	ESOP list	one Toffoli/cube	NCT QC	—
2016	Existing [8]	ESOP list	UMG/UNG grid	GA, QC	—
2016	Existing [14]	ESOP list	sorted sub-products	GA, GB, QC	CMOS
2017	Existing [16]	2^n / LUT	functional hierarchy	NCT	—
2024	Existing [13]	modulo 2^n+1	QMA1–QMA4 adders	D_{CCX}, FOM	Qiskit
2026	Existing [20]	LNN DQC	swap vs. switch	gates, X, C	Qiskit
—	**This work**	fixed ESOP	**Prefix Cube tree**	GA; G, D, Q	PTL+CNT+Qiskit

three fabrics; Section V reports the MCNC results and the experimental workflow of Section V-A.

II. BACKGROUND STUDY

A. Related Work

Mitra and Rahman [8] introduced UMG/UNG and a pairwise grid that reuses garbage as later AND inputs. Tara *et al.* [14] sort the product list and share sub-products with an NMG (qc=4); that sort does not build a prefix tree. Transformation-based synthesis [15] and hierarchical LUT synthesis [16] start from a 2^n table and emit an NCT cascade [17]. An ESOP Toffoli cascade [18] rebuilds every cube. With n up to 65, that table is a different problem. RevLib catalogs the NCT style [19]. Zhou *et al.* [20] compile onto a linear nearest-neighbor line and, for each interaction, pick the cheaper of a SWAP path, a switch, and gate teleportation. Section V-D applies that choice, the NCT cascade of [12], and the QMA3 zero-reset rule of [13] to the same cover P. Table II is the comparison class.

B. Motivation of Work

Each rebuilt prefix is another UMG/UNG and another ancilla, and (2) charges it $\mathrm{qc} = 5$. Prefix Cube cuts n_{AND} on the fixed cover P before technology mapping. The mapper is one sharing tree that realizes (3), counts one AND per distinct multi-literal prefix, and is read as Nangate45 PTL, Stanford CNTFET/VTEAM, and a Qiskit circuit scored by G, D, and Q. Readout is a computational-basis measurement of the Bennett outputs. Surface-code correction is outside this CAD [21], [22].

C. Contributions

The main contributions of this paper are as follows.

- $C_\wedge = \mathrm{AI}_{\mathrm{AND}} = \frac{1}{5}\mathrm{QC}_{\mathrm{AND}} = |V_{\geq 2}|$ (Lemma 1, Theorem 1).
- Algorithms 1–3 with the published Feynman XOR plane, in $O(n^2p + pk + pm)$ time and independent of 2^n (Theorem 3), including uncomputation and computational-basis measurement.
- One netlist, $\mathcal{N}_{\mathrm{PTL}} \equiv \mathcal{N}_{\mathrm{CNT}} \equiv U_{\mathrm{RPLA}} = \mathcal{N}$ (Theorem 4), read as Nangate45 PTL, Stanford CNTFET/VTEAM, and Qiskit. Macro cost remains CV^2.
- On 20 representative MCNC PLA circuits, mean GA/AI/QC fall by 53.4/62.3/53.2% versus [14]. After `qiskit-transpile`, mean NCT G, D, and Q are 6492.2, 907.5, and 349.6; Prefix Cube reaches G = 1154.0 (82.2%), an AND-plane depth bound of 22.3, and $Q = 785.6$ (Table X). Median runtime is 9.4 ms.

On MCNC 'misex1' (20 products), Prefix Cube uses $|V_{\geq 2}| = 17$ (16 UMG, 1 UNG) plus 28 Feynman XORs; the same $\mathcal{N}$ is reused in Figs. 2–6.

III. PREFIX CUBE BASED AND PLANE

Prefix Cube synthesizes $T(P, \sigma)$ so that (1) holds. The XOR plane is ConstructXORPlane (Algorithm 5 of [8]).

A. Prefix Cubes

Let $X = \{x_1, \ldots, x_n\}$ and let a cube c have literals on distinct variables. A permutation σ of X orders the literals of each cube as c^σ.

Definition 1 (Prefix Cube Tree). *For* $c^\sigma = \ell_1 \cdots \ell_k$, $\mathrm{pre}_r(c^\sigma) = \ell_1 \cdots \ell_r$. *The family is* $\Pi(P, \sigma) = \bigcup_{c \in P} \{\mathrm{pre}_r(c^\sigma) : 1 \leq r \leq |c|\}$. *The tree* $T(P, \sigma) = (V, E)$ *has* $V = \{\varepsilon\} \cup \Pi(P, \sigma)$ *and an edge* $\mathrm{pre}_{r-1} \to \mathrm{pre}_r$ *for every cube and* $1 \leq r \leq |c|$. *Depth-1 nodes are primary rails. Depth-*≥ 2 *nodes are AND results. Write* $V_{\geq 2} = \{\pi \in \Pi : |\pi| \geq 2\}$.

Lemma 1 (Prefix AND Count). *Any AND plane that realizes every* $c \in P$ *from two-input ANDs along* σ *uses at least* $|V_{\geq 2}|$ *ANDs. The Prefix Cube tree attains* (1).

Proof: Distinct width-≥ 2 prefixes cannot share an AND output. Creating each $\pi \in V_{\geq 2}$ from its parent by one AND yields every leaf. A shared prefix of length at least 2 is one node of $V_{\geq 2}$. ∎

The objective $\sigma^\star \in \arg\min_\sigma |V_{\geq 2}|$ is combinatorial. Algorithm 1 is a polarity-aware greedy heuristic, not an approximation algorithm.

B. Prefix Cube Construction Algorithms

Algorithm 1 appends the variable that appears in the most current prefix buckets, then splits buckets by polarity. Algorithm 2 inserts each ordered cube into a trie. Each *new* depth-≥ 2 edge is one OpAND (Algorithm 1 of [8]). OpAND is a published 3×3 cell of [8]: UMG is the bijection $(a, b, c) \mapsto (a,\ b \oplus c,\ ab \oplus \bar{a}c)$ with $\mathrm{qc} = 5$ and $r|_{c=0} = a \wedge b$ while rail a survives. UNG is $(a, b, c) \mapsto (a, b, (a \vee b) \oplus c)$ with $\mathrm{qc} = 5$ and covers $\bar{a}\bar{b}$ by De Morgan. Templates $\{\alpha, \beta, \gamma, \pi\}$ realize ab, $a\bar{b}$, $\bar{a}b$, $\bar{a}\bar{b}$. Primed variants $\{\alpha', \beta', \gamma', \pi'\}$ swap rails [8]. A filled crosspoint instantiates a UMG/UNG the first time a prefix appears. An open tap reuses it.

Algorithm 1 SelectPrefixOrder(P)

Require: Product set P over X
Ensure: Literal order σ
1: $\sigma \leftarrow (\,)$; $U \leftarrow X$; $\mathcal{B} \leftarrow \{P\}$
2: **while** $U \neq \emptyset$ **do**
3: $x^\star \leftarrow \arg\max_{x \in U} \sum_{B \in \mathcal{B}} |\{c \in B : x \in \mathrm{vars}(c)\}|$
4: Append $x^\star$ to σ; $U \leftarrow U \setminus \{x^\star\}$
5: Split each bucket by polarity of $x^\star$ (positive, negative, omit)
6: **end while**
7: **return** σ

Algorithm 2 BuildPrefixCubeTree(P, σ)

Require: P, order σ
Ensure: Tree T, product map λ, AND netlist
1: $V \leftarrow \{\varepsilon\}$; $M[\varepsilon] \leftarrow$ (no wire)
2: **for all** $c \in P$ **do**
3: $\pi \leftarrow \varepsilon$
4: **for all** literal ℓ of c^σ **do**
5: **if** $(\pi \cdot \ell) \notin V$ **then**
6: add node $\pi \cdot \ell$
7: **if** $|\pi \cdot \ell| = 1$ **then** $M[\pi \cdot \ell] \leftarrow$ rail of ℓ
8: **else** $M[\pi \cdot \ell] \leftarrow \mathrm{OpAND}(M[\pi], \ell)$
9: **end if**
10: **end if**
11: $\pi \leftarrow \pi \cdot \ell$
12: **end for**
13: $\lambda(c) \leftarrow \pi$
14: **end for**

Proposition 1. *Let $p = |P|$, $k = \max|c|$, $n = |X|$. Algorithm 1 is $O(n^2p)$. Algorithm 2 is $O(pk)$ time and $O(|V|)$ memory with $|V| \le 1+np$. Neither searches 2^n minterms nor $n!$ orders.*

Fig. 1 is that tree on $P = \{ab', ab'c, ac, a'b'c, bc'\}$ under $\sigma = (a, b, c)$. The extend edge $ab' \to ab'c$ reuses ab', and $|V_{\ge 2}| = 6$. Every teal or dashed-orange edge is one UMG/UNG of $\mathrm{qc} = 5$.

Lemma 2 (Template Completeness)**.** *Templates $\{\alpha, \beta, \gamma, \pi\}$ realize every two-literal polarity. Primed variants cover swapped rails. Hence every Prefix Cube edge is realizable by one OpAND of [8].*

Theorem 1 (AND Plane)**.** *Algorithm 2 realizes every $c \in P$ bijectively, with $G_{\mathrm{AND}} = \mathrm{QC}_{\mathrm{AND}}/5 = \mathrm{AI}_{\mathrm{AND}} = |V_{\ge 2}|$.*

Proof: Cubes follow a unique σ-path from the root, so a parent wire exists before each child. OpAND is a permutation. Leaves are the products. Lemma 1 counts ANDs. ∎

Theorem 2 (XOR-Plane Resources [8])**.** *ConstructXORPlane of [8] applied to Prefix Cube leaves is correct. With n XOR operations, m outputs, and $xdot$ Feynman cross-points, $G_{\mathrm{XOR}} = \mathrm{QC}_{\mathrm{XOR}} = n + m - xdot$ and $\mathrm{AI}_{\mathrm{XOR}} = m - xdot$. Combined with Theorem 1,*

$$\mathrm{QC}(\mathcal{N}) = 5|V_{\ge 2}| + (n + m - xdot). \tag{5}$$

The XOR plane uses Feynman templates △ (NOT), λ (XOR), and ▽ (COPY) [7], [8].

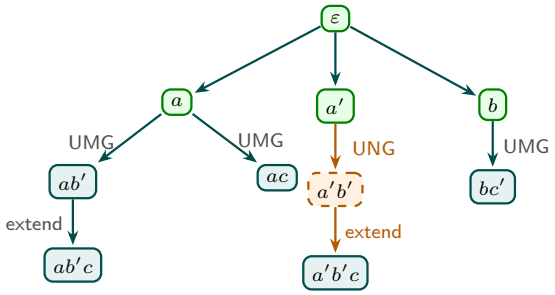


Fig. 1. Example Prefix Cube tree of Algorithm 2 for $P = \{ab', ab'c, ac, a'b'c, bc'\}$ under $\sigma = (a, b, c)$. Green: depth-1 rails. Teal: UMG OpAND. Dashed orange: UNG. The edge $ab' \to ab'c$ reuses ab' ($|V_{\ge 2}| = 6$).

Theorem 3 (CAD Complexity)**.** *SelectPrefixOrder, BuildPrefixCubeTree, and ConstructXORPlane run in $O(n^2p+pk+pm)$ time and $O(|V|+m)$ memory, polynomial in the ESOP listing and independent of 2^n.*

Proof: Proposition 1 gives $O(n^2p+pk)$ for σ and T. ConstructXORPlane scans products against output membership, which is $O(pm)$. The netlist stores one gate per depth-≥ 2 edge plus XOR macros. ∎

Nangate45 PTL budgets (Table III) follow Zhou *et al.* [23]: UMG is an 8T XOR plus 4T MUX plus restorers and a sleep footer (17T). UNG is a 4T NOR plus XOR (23T). Feynman is 11T. AND-plane transistor count is $17|V_{\ge 2}^{\mathrm{UMG}}| + 23|V_{\ge 2}^{\mathrm{UNG}}|$. Logical reversibility of the truth table does not recover the CV^2 of restored rails.

Table III is the PTL interpretation of each cell in $\mathcal{N}$, not a second synthesis. UMG is 17T, UNG 23T, and Feynman 11T, including restore and a shared high-V_t sleep footer. AND-plane transistor count is therefore $17|V_{\ge 2}^{\mathrm{UMG}}| + 23|V_{\ge 2}^{\mathrm{UNG}}|$. Prefix Cube lowers N_{T} only by cutting $|V_{\ge 2}|$, not by changing the Zhou budgets.

C. Running Example: MCNC `misex1`

MCNC `misex1` has n=8 inputs, m=7 outputs, and 20 unique ESOP products after duplicate rows are merged [6]. Under $\sigma = (b, c, a, d, f, e, g, h)$, Fig. 2 has $|V_{\ge 2}| = 17$, so $G_{\mathrm{AND}} = 17$ and $\mathrm{QC}_{\mathrm{AND}} = 85$. Shared prefixes $b'c$ (eight cubes) and bc' (five cubes) are generated once (Table IV). The specification is

$$F = \begin{cases} f_1 = P_1 \oplus P_2, \\ f_2 = P_3 \oplus P_4 \oplus P_5 \oplus P_6 \oplus P_7, \\ f_3 = P_5 \oplus P_8 \oplus P_9 \oplus P_{10} \oplus P_{11}, \\ f_4 = P_2 \oplus P_{12} \oplus P_{13} \oplus P_{14}, \\ f_5 = P_3 \oplus P_5 \oplus P_8 \oplus P_{15} \oplus P_{16} \oplus P_{17}, \\ f_6 = P_2 \oplus P_3 \oplus P_5 \oplus P_6 \oplus P_{18} \oplus P_{19}, \\ f_7 = P_2 \oplus P_5 \oplus P_8 \oplus P_{13} \oplus P_{20}. \end{cases} \tag{6}$$

The XOR plane is ConstructXORPlane on those twenty leaves.

Fig. 2 is Algorithm 2 on these twenty cubes. SelectPrefixOrder fixes $\sigma = (b, c, a, d, f, e, g, h)$, and each cube is written in that variable order with its polarity kept. The trie starts at ε. The first literal of a cube is a depth-1 rail, drawn

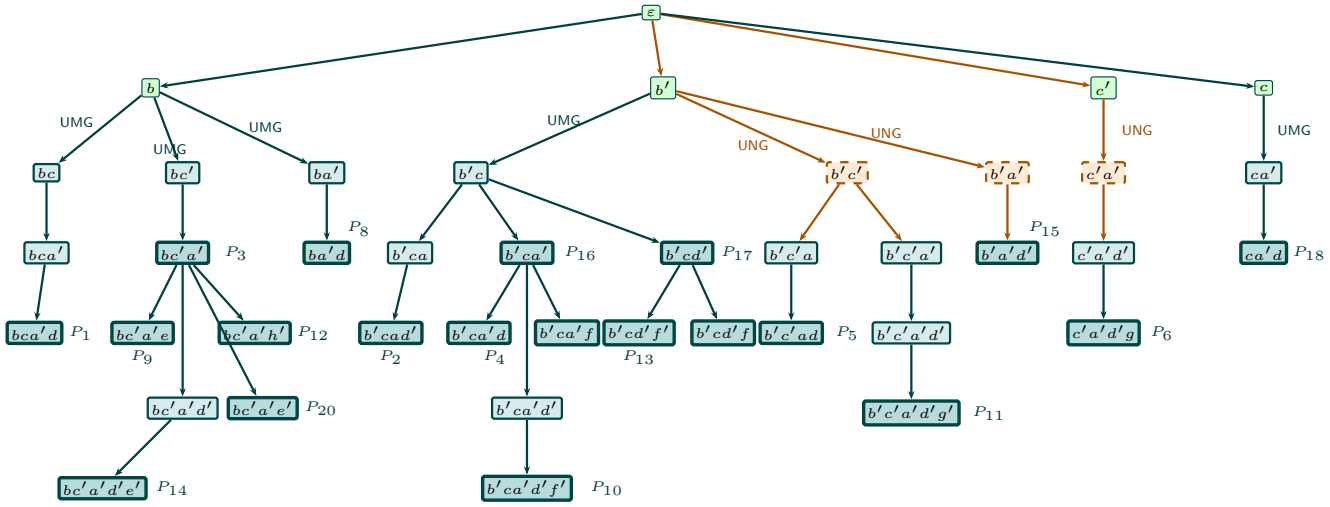


Fig. 2. Prefix Cube tree $T(P,\sigma)$ for `misex1` with $\sigma=(b,c,a,d,f,e,g,h)$. Thick nodes are ESOP products $P_1,\ldots,P_{20}$. The edge $b'c \to b'ca'$ is reused by four products. Green: depth-1 rails. Teal fill: Prefix Cube (UMG). Dashed orange: UNG ($b'c'$, $b'a'$, $c'a'$). Shared $b'c$ (8 cubes) and bc' (5 cubes) are reused, not rebuilt.

TABLE III
NANGATE45 PTL TRANSISTOR BUDGET

Macro	Core	Restore+SL	Total
UMG	8T XOR + 4T MUX + 2T $\bar{a}$	3	17
UNG	4T NOR + 2T INV + 8T XOR	9	23
Feynman	8T XOR	3	11

TABLE IV
TWO-INPUT AND COUNT FOR `misex1`

AND-plane method	$C_\wedge$	Shared width-≥ 2 prefixes
Existing [14]	product list	heuristic list sharing
Prefix Cube tree (Alg. 2)	17	$b'c$, $bc'a'$, $b'ca'$, $b'cd'$

green: b, b', c, and c'. Those rails are primary inputs. Each later literal either meets a prefix that already exists, and then takes an open tap, or creates a new prefix, and then inserts one OpAND from the parent wire. A teal node is a UMG. A dashed orange node is a UNG, drawn for the complemented prefixes $b'c'$, $b'a'$, and $c'a'$. A thick border marks an ESOP product; when that product is also a prefix of a longer cube, later cubes tap it.

The prefix $b'c$ is inserted once and then tapped by eight cubes: $P_2{=}b'cad'$, $P_{16}{=}b'ca'$, $P_4{=}b'ca'd$, $P_7{=}b'ca'f$, $P_{10}{=}b'ca'd'f'$, $P_{17}{=}b'cd'$, $P_{13}{=}b'cd'f'$, and $P_{19}{=}b'cd'f$. The edge $b'c \to b'ca'$ serves P_{16}, P_4, P_7, and P_{10}; from $P_{17}{=}b'cd'$ the tree continues to P_{13} and P_{19}. The prefix bc' is likewise inserted once: $P_3{=}bc'a'$ is the common node of $P_9{=}bc'a'e$, $P_{12}{=}bc'a'h'$, $P_{14}{=}bc'a'd'e'$, and $P_{20}{=}bc'a'e'$. The other products occupy shorter branches: $P_1{=}bca'd$ and $P_8{=}ba'd$ under b; $P_5{=}b'c'ad$, $P_{11}{=}b'c'a'd'g'$, and $P_{15}{=}b'a'd'$ under b'; $P_6{=}c'a'd'g$ under c'; and $P_{18}{=}ca'd$ under c. ConstructXORPlane then combines these twenty leaves into $f_1,\ldots,f_7$ by Feynman gates (6). Every new width-≥ 2 prefix is one cell, so the AND plane of this trie has $|V_{\geq 2}| = 17$.

Table IV records that count: [14] shares sub-products on a product list; Prefix Cube attains $C_\wedge = 17$.

The Prefix Cube tree is the programmable AND plane of a full reversible PLA grid (Fig. 3).

Fig. 3 places that AND plane on a programmable grid: 16 UMG and 1 UNG ($|V_{\geq 2}| = 17$) plus 28 Feynman XOR cells, OpenROAD pitch $(n, |V_{\geq 2}|) = (8, 17)$.

IV. SAME NETLIST, THREE FABRICS

The experimental workflow of Section V-A is as follows. A fixed ESOP cover P (classic `.esop`, ABC/EXORCISM-4, or published `.pla`) is synthesized by Existing [14] and by Prefix Cube—then one netlist $\mathcal{N}$ is *interpreted*, not resynthesized, as Nangate45 PTL, Stanford CNTFET/VTEAM, and Qiskit. Logic counters GA, GB, AI, and QC are properties of $\mathcal{N}$. Area, wirelength, power, and delay are OpenROAD/OpenSTA/ngspice of that same $\mathcal{N}$. The four output paradigms (non-reversible PLA, reversible PLA, reversible post-CMOS PLA, and the Qiskit embedding) share the cover. Only the last three realize (3).

Theorem 4 (Cross-Fabric Identity)**.** *$\mathcal{N}$ of Algorithms 1–2 plus ConstructXORPlane is fabric-independent:*

$$\mathcal{N}_{\mathrm{PTL}} \equiv \mathcal{N}_{\mathrm{CNT}} \equiv U_{\mathrm{RPLA}} = \mathcal{N}. \tag{7}$$

Thus $|V_{\geq 2}|$, GA, AI, and QC are invariant. Area, power, and delay units change. PTL and VTEAM maps dissipate

$$P_{\mathrm{dyn}} \propto N_{\mathrm{T}} C_{\mathrm{sw}} V_{\mathrm{DD}}^2 f, \qquad P_{\mathrm{leak}} \propto I_{\mathrm{off}} V_{\mathrm{DD}} N_{\mathrm{T}} \tag{8}$$

and follow ordinary CV^2 switching [24]–[26].

A. Logic Synthesis

The Boolean branch of the workflow is as follows. Algorithms 1–2 insert each cube of P into $T(P,\sigma)$ and emit one OpAND per depth-≥ 2 edge. ConstructXORPlane of [8] is the EX-OR plane. Figure 1 is that tree on the five-cube cover $\{ab', ab'c, ac, a'b'c, bc'\}$ ($|V_{\geq 2}| = 6$). Figs. 2 and 4 are the same construction on `misex1`: twenty ESOP products, shared

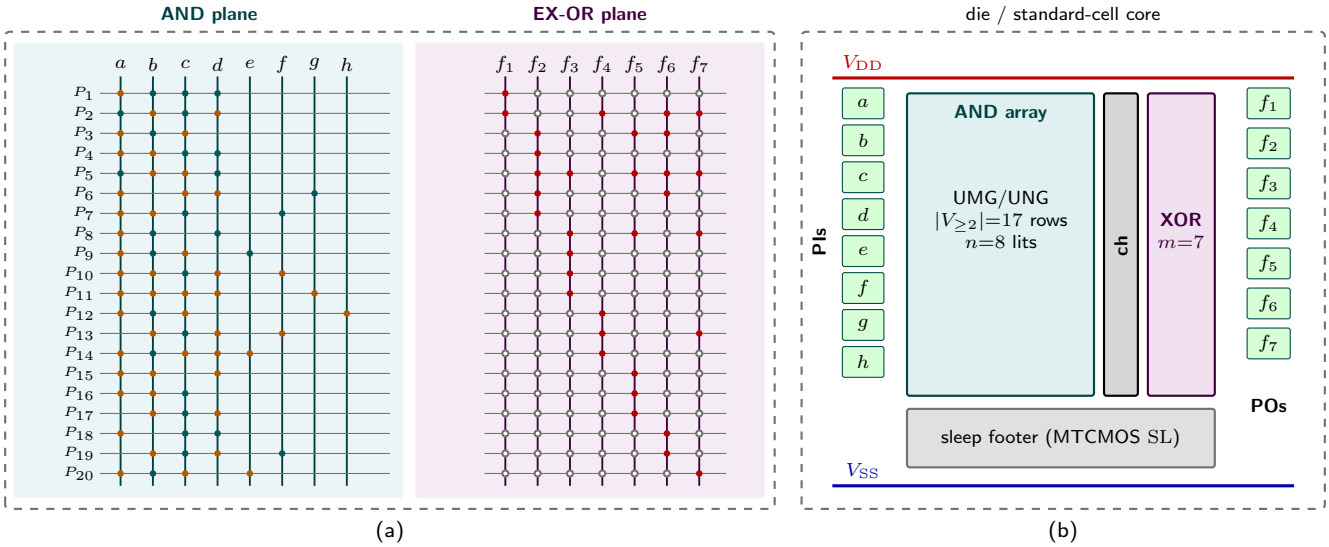


Fig. 3. Programmable RPLA of `misex1`. (a) RPLA grid of MCNC `misex1` (8 in / 20 products / 7 out). Filled teal marks literal 1. Filled orange marks literal 0. Open sites are don't-care / HRS. Violet fill marks XOR incidence. The AND macros are 16 UMG and 1 UNG ($|V_{\geq 2}|$=17). There are 28 Feynman XOR cells. (b) Row-based placement of pitch $(n, |V_{\geq 2}|) = (8, 17)$. OpenROAD rows of PTL macros for `misex1`.

prefixes $b'c$ (eight cubes) and $bc'a'$ (five cubes), $|V_{\geq 2}| = 17$, and 28 Feynman combining cells (Table IV). UMG, UNG, and Feynman are the cell library (qc 5, 5, and 1). NMG appears in [14]. The mapper does not allocate a 2^n table (Theorem 3).

Fig. 4 is the same netlist as an RPLA schematic: shared prefixes $b'c$ and $bc'a'$, $|V_{\geq 2}| = 17$, and 28 Feynman cells into (6).

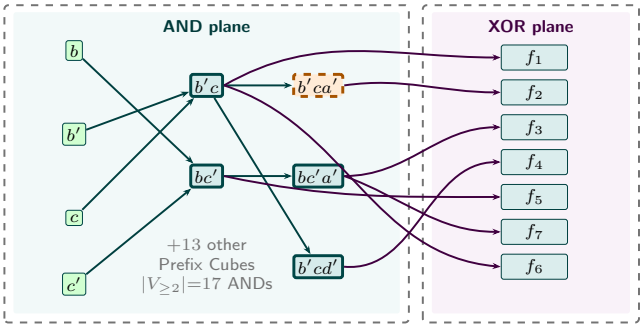


Fig. 4. Prefix Cube RPLA of `misex1`: AND-plane tree (shared prefixes $b'c$, $bc'a'$, $|V_{\geq 2}| = 17$) and Feynman ESOP combining. Shared $b'c$ (8 cubes), $bc'a'$ (5 cubes).

B. Technologies Mapping

Technology mapping interprets that same $\mathcal{N}$. *PTL.* Yosys, OpenROAD, OpenSTA, and ngspice under Nangate45 instantiate the Zhou UMG/UNG/FG budgets of Table III. Table V reports the corresponding OpenROAD/OpenSTA/ngspice area, wirelength, average power, and t_{pLH}/t_{pHL} of those macros versus Feynman, Toffoli, Fredkin, Peres, MUX, and NMG [14], each on PTL, CNTFET, and VTEAM. Figure 3(a) is the programmable AND–XOR grid of `misex1` (8 in / 20 products / 7 out). Panel (b) is the OpenROAD row placement of pitch $(n, |V_{\geq 2}|) = (8, 17)$. The programmed array uses

$$N_{\mathrm{T}}(F) = 16 \cdot 17 + 23 + 28 \cdot 11 = 603 \tag{9}$$

MOS transistors plus the distributed sleep row. *Post-CMOS.* Stanford 32 nm VS-CNFET and VTEAM 1T1R interpret the same 3×3 I/O at $V_{\mathrm{DD}} = 0.9\,\mathrm{V}$ (Fig. 5): complementary wrap-gate UMG channels in panel (a) and LRS/HRS 1T1R UNG in panel (b). That map is $\mathcal{N}_{\mathrm{CNT}}(F) = \mathcal{N}_{\mathrm{mem}}(F) = \mathcal{N}(F)$. Prefix Cube taps add no extra 1R. With AND-plane program bits $\Omega_{i,k}$ and XOR-plane bits $\Lambda_{i,j}$,

$$N_{\mathrm{LRS}}(F) = \sum_{i,k} \Omega_{i,k} + \sum_{i,j} \Lambda_{i,j} = 44 + 33 = 77. \tag{10}$$

Table V and Fig. 5 are the PTL and post-CMOS interpretations of $\mathcal{N}$, not a second synthesis. Nangate45 cells are essentially iso-area (63–67 $\mu\mathrm{m}^2$), so RPLA area and power track how many AND macros are instantiated—the Prefix Cube objective. CNTFET shrinks UMG area by about 300× and conventional switching power by about 187× at $V_{\mathrm{DD}} = 0.9\,\mathrm{V}$. The Qiskit embedding does not use these milliwatts; its ledger is G, D, and Q. Milliwatts remain CV^2 of N_{T} devices (8), not recovered Landauer energy.

C. Quantum Circuit Design

This subsection is the Qiskit embedding of the same netlist, not a quantum algorithm. Table I applies: GA/QC count macros on $\mathcal{N}$; G, D, and Q count operations, depth, and qubits on a Qiskit circuit [1]. A rail of $\mathcal{N}$ becomes a qubit only after Algorithm 3.

a) Algorithm.: The host CAD is Algorithms 1–2 plus ConstructXORPlane of [8]. It emits $\mathcal{N}$ in $O(n^2p + pk + pm)$ time, independent of 2^n (Theorem 3). Algorithm 3 interprets that netlist as the computational-basis unitary U_{RPLA} of (4) [9], [10], [17]. Each UMG/UNG is one five-cost AND macro and each Feynman COPY/XOR is one CNOT, so the macro cost remains (5). The comparison flow is the ESOP Toffoli cascade of [12], [18] after qiskit-transpile

TABLE V
DEVICE-LEVEL COST OF THE PROPOSED UMG AND UNG MACROS VERSUS STANDARD REVERSIBLE GATES, INCLUDING NMG [14]. UNDER EACH METRIC THE THREE COLUMNS ARE PTL$^{\tau}$ (NANGATE45 YOSYS/OPENROAD/OPENSTA/NGSPICE), CNTFET$^{\xi}$ (32 NM STANFORD VS-CNFET, NGSPICE), AND MEMRISTOR$^{\eta}$ (VTEAM, NGSPICE). AREA AND WIRELENGTH ARE OPENROAD; AVERAGE POWER AND t_{pLH}/t_{pHL} ARE OPENSTA/NGSPICE. THESE CELLS ARE THE TECHNOLOGY MAP OF THE REVERSIBLE PLA AND REVERSIBLE POST-CMOS PLA PARADIGMS.

Gate	Area (μm^2)			WL (μm)			Avg. P (10^{-4}W)			t_{pLH} (ps)			t_{pHL} (ps)		
	PTL$^{\tau}$	CNTFET$^{\xi}$	memristor$^{\eta}$	PTL$^{\tau}$	CNTFET$^{\xi}$	memristor$^{\eta}$	PTL$^{\tau}$	CNTFET$^{\xi}$	memristor$^{\eta}$	PTL$^{\tau}$	CNTFET$^{\xi}$	memristor$^{\eta}$	PTL$^{\tau}$	CNTFET$^{\xi}$	memristor$^{\eta}$
Feynman [7]	63	0.0861	0.3707	42	0.2737	0.8450	1.076	0.007	3.869	7297	7444	7796	19874	19418	19796
Toffoli [27]	64	0.1715	0.5687	179	0.5206	1.0725	1.355	0.014	6.476	7664	19373	19795	7695	64395	66596
Fredkin [28]	64	0.2580	0.6321	342	0.8077	1.5600	1.877	0.015	12.15	6255	16697	16205	8321	16323	18596
Peres [29]	65	0.1722	0.5814	74	0.5474	1.1700	2.272	0.017	7.781	7664	19373	19796	7695	64395	66596
MUX [30]	66	0.3007	0.6448	313	0.9311	1.6575	3.300	0.018	12.22	7664	19373	19796	43495	64395	66596
NMG [14]	67	0.3437	0.6701	102	1.0679	1.8525	2.71261	0.01454	13.99	7707.7	7374	7795.0	18365.4	16392.9	18596.2
UMG [8]	65	0.2153	0.6068	210	0.6842	1.3650	2.426	0.013	9.624	7664	19373	19795	43470	64395	66596
UNG [8]	64	0.1722	0.5814	179	0.5474	1.1700	1.645	0.011	7.685	4571	4557	4204	79339	79349	79795

$^{\tau}$Nangate45 (Yosys/OpenROAD/OpenSTA/ngspice); $^{\xi}$32 nm CNFET (ngspice); $^{\eta}$VTEAM memristor (ngspice).

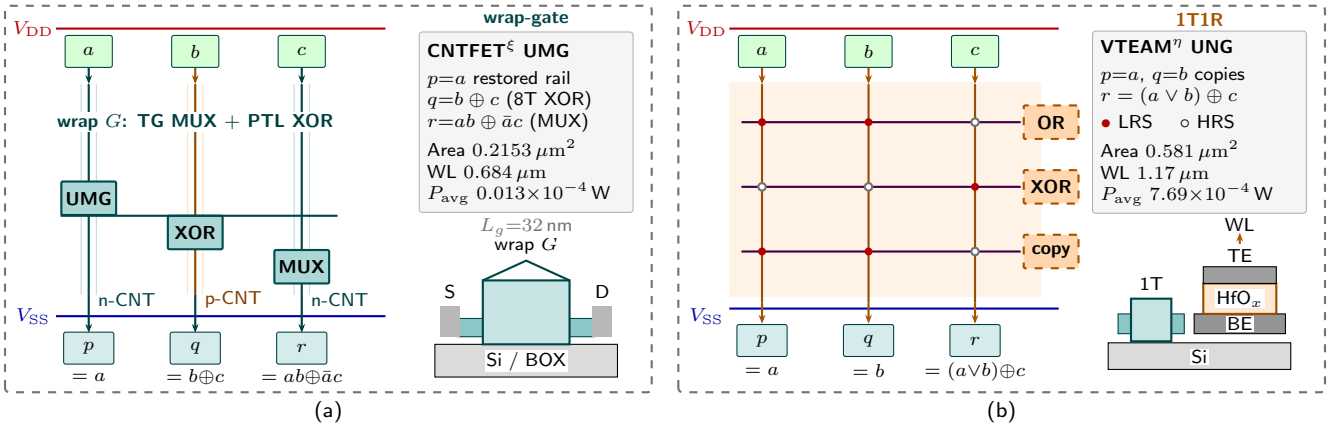


Fig. 5. Post-CMOS 3×3 macros. (a) Stanford VS-CNFET UMG: complementary n-CNT/p-CNT channels, wrap-gate MUX/XOR, and a 32 nm wrap-gate stack. (b) VTEAM 1T1R UNG: LRS/HRS program, OR/XOR/copy wordlines, and a TiN/HfO$_x$ cell. Milliwatts are conventional CV^2, not recovered Landauer energy.

(`optimization_level=2`, `seed_transpiler=42`). That flow is scored by gate count G, depth D, and qubit count Q (Table X). Algorithm 4 applies the same transpiler to the circuit of Algorithm 3, keeping each UMG/UNG macro in the basis. Gaur's resource vector and Zhou's swap-versus-switch cost are additional scores of the same NCT circuit [13], [20].

Algorithm 4 is the Qiskit 2.3.1 pass on the circuit of Algorithm 3. It leaves P and $\mathcal{N}$ unchanged. Circuits above 96 qubits or 8000 gates stay logical. The reported Prefix Cube gate count remains the UMG/UNG/CX count of $\mathcal{N}$, and $D_{\wedge} \le n-1$ remains the tree-height bound.

Lemma 3 (Embedding Width)**.** *Algorithm 3 returns a circuit on*

$$Q = n + \mathrm{AI} = n + |V_{\ge 2}| + (m - xdot) \tag{11}$$

qubits.

Proof: The algorithm sets the register width to $Q = n + \mathrm{AI}$. Theorem 1 gives one prefix ancilla per node of $V_{\ge 2}$. Theorem 2 gives $\mathrm{AI}_{\mathrm{XOR}} = m - xdot$. The sum is (11). ■

Proposition 2 (AND-Plane Depth)**.** *In the topological schedule of Algorithm 3, the AND plane has depth $D_{\wedge} \le n-1$.*

Algorithm 3 EmbedPrefixCubeQiskit($\mathcal{N}$)

Require: Prefix Cube netlist $\mathcal{N}$ with tree $T(P, \sigma)$ and Feynman XOR plane
Ensure: Qiskit circuit, then m computational-basis bits
1: Allocate $Q = n + \mathrm{AI}$ qubits: inputs x, prefix ancilla a, outputs y
2: Initialize $a \leftarrow |0\rangle^{\otimes |V_{\ge 2}|}$, $y \leftarrow |0\rangle^{\otimes m}$
3: **for all** depth-d prefix $\pi \in V_{\ge 2}$ in topological order **do**
4: apply one UMG/UNG of [8] to parent(π) and literal(π)
5: **end for**
6: **for all** XOR incidence (c, f_j) **do**
7: $\mathrm{CX}(\lambda(c), y_j)$ ▷ Feynman combining
8: **end for**
9: apply the inverse AND schedule $U_{\mathrm{AND}}^{\dagger}$ to restore a to $|0\rangle$
10: measure y in the computational basis; leave x unmeasured
11: **return** the `QuantumCircuit` Φ

Proof: A cube contains at most n literals. Its first literal is a depth-1 rail, so the OpAND chain has length at most $n-1$. Prefixes of equal depth have no path between them and share one layer. ■

Theorem 5 (Computational-Basis Readout)**.** *On input $|x\rangle|0\rangle$,*

Algorithm 4 TranspilePrefixCubeQiskit(Φ)

Require: `QuantumCircuit` Φ from Algorithm 3
Ensure: Transpiled circuit and its (G, D, Q)
1: $Q \leftarrow$ qubits of Φ; $G_0 \leftarrow$ gates of Φ
2: **if** $Q > 96$ or $G_0 > 8000$ **then**
3: **return** Φ ▷ logical circuit; transpile skipped
4: **end if**
5: Basis $\mathcal{B} \leftarrow \{x, \mathrm{cx}, \mathrm{umg}, \mathrm{ung}, \mathrm{swap}\}$
6: $\Phi' \leftarrow$qiskit-transpile($\Phi, \mathcal{B}$, `optimization_level`= 2, `seed`= 42)
7: $G \leftarrow$ gates of Φ'; $D \leftarrow \mathrm{depth}(\Phi')$; $Q \leftarrow$ qubits of Φ'
8: **return** (Φ', G, D, Q)

Algorithm 3 realizes the Bennett embedding (3): $U_{\mathrm{AND}}^{\dagger}$ restores every prefix ancilla to $|0\rangle$, and a computational-basis measurement of y returns $f(x)$.

Proof: Theorem 1 makes U_{AND} a permutation with a parent wire before each child, so the inverse schedule restores a. Each Feynman CX of ConstructXORPlane writes one product parity onto an output rail and leaves the product wire in place. The algorithm measures those m output qubits and leaves the input rails unmeasured. ∎

Proposition 3 (Transpile Ledger)**.** *Algorithm 4 leaves P and $\mathcal{N}$ unchanged. Above 96 qubits or 8000 gates it returns the logical circuit. The Prefix Cube ledger keeps G equal to the UMG/UNG/CX count of $\mathcal{N}$ and keeps $D_{\wedge} \leq n-1$.*

Proof: The algorithm reads Φ and either returns it or calls qiskit-transpile on that circuit. The cover and the netlist are not inputs of the call. Reported G is taken from $\mathcal{N}$, and $D_{\wedge}$ is the bound of Proposition 2. ∎

Theorem 6 (Shared UMG/UNG Gap)**.** *With* $\mathrm{qc(UMG)} = \mathrm{qc(UNG)} = \mathrm{qc(T)} = 5$ *and every order σ,*

$$|V_{\geq 2}| \leq n_{\mathrm{CCX}} = \sum_{c \in P} \max(|c| - 1, 0), \tag{12}$$

so the AND-plane gap $5(n_{\mathrm{CCX}} - |V_{\geq 2}|)$ is nonnegative and is positive when a width-≥ 2 prefix lies on two cubes. QMA3 [13] keeps the NCT QC. Algorithm 1 of [20] adds $N_{\mathrm{link}} \geq 0$, so its QC is at least that NCT QC.

Proof: A cube of ℓ literals contributes at most $\ell - 1$ nodes of $V_{\geq 2}$, and a shared node is counted once, which is (12). The NCT AND plane [12], [18] adds $|c| - 1$ per cube. Theorem 1 charges the trie $5|V_{\geq 2}|$. QMA3 keeps the Toffoli count. Each Zhou link adds one gate. ∎

On `misex1`, $b'c$ and bc' make (12) strict: the AND plane costs $5 \cdot 17 = 85$ inside QC $= 113$, against NCT QC 319. Mean QC on the 20 covers falls by 69.0% versus NCT and Gaur, and by 74.8% versus Zhou. Mean GA/AI/QC fall by 53.4/62.3/53.2% versus [14]. Depth stays $D_{\wedge} \leq n - 1$. Each saved Toffoli still occupies a rail in (11), so Q can exceed a cascade with fewer ancilla lines.

Figure 6 is Algorithm 3 on `misex1`. Panel (a) is the NCT Toffoli cascade: twenty open-control ANDs plus Feynman XOR into $f_1, \ldots, f_7$ (91 reversible gates, QC $= 319$; Qiskit G=134, D=46, Q=38). Panel (b) is Prefix Cube: $|V_{\geq 2}| = 17$ UMG/UNG macros on shared prefixes $b'c$ and $bc'a'$, then the same XOR plane (G_{PC}=45, QC=113, $Q = n +$ AI=27, AND-plane depth at most $n-1$=7). The two panels share the wire order. Only the AND-plane cells change. Section V-D scores that comparison on 20 representative MCNC PLA circuits.

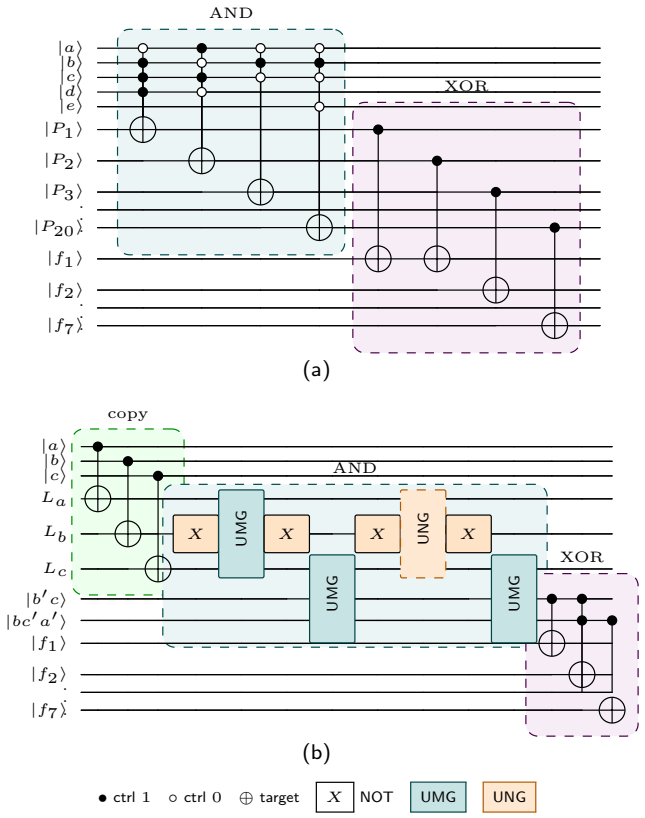


Fig. 6. Qiskit embeddings of the same `misex1` netlist $\mathcal{N}$ (Algorithm 3). (a) Existing NCT Toffoli cascade [12]: G=134, D=46, Q=38 after transpile (91 reversible gates, QC = 319). (b) Prefix Cube UMG/UNG on shared prefixes $b'c$ and $bc'a'$ ($|V_{\geq 2}| = 17$): G=45, $D_{\wedge} \leq 7$, Q=27 (QC = 113). The panels share the wire order; only the AND-plane cells change.

The function f is classical: running Φ of (3) on qubits does not reduce $|P|$ or yield a query speedup [10]. The advantage is fewer AND macros than product-list sorts and NCT cascades on the same cover, and Qiskit is a ledger for G, D, and Q (Table X), not an oracle. Algorithm 3 uncomputes prefix ancillae by $U_{\mathrm{AND}}^{\dagger}$ and measures the m Bennett outputs in the computational basis; surface-code overhead is outside this CAD, and no noisy or fault-tolerant speedup is claimed [21], [22]. Prefix Cube lowers G by sharing prefixes and can raise Q because each distinct prefix occupies a rail.

V. Experimental Evaluation

Prefix Cube is scored on 20 representative MCNC PLA circuits [6]. The experimental covers were retrieved from the public MCNC/EPFL HDL benchmark archive; the Ubuntu host on which the simulated flow ran, the Nangate45, Stanford CNFET, and VTEAM libraries used to extract the metrics, and the tool versions are listed in Appendix A.

A. Protocol

Fig. 7 evaluates one fixed ESOP cover P: a published `.esop`, otherwise ABC/EXORCISM-4 (Q=2), or a published

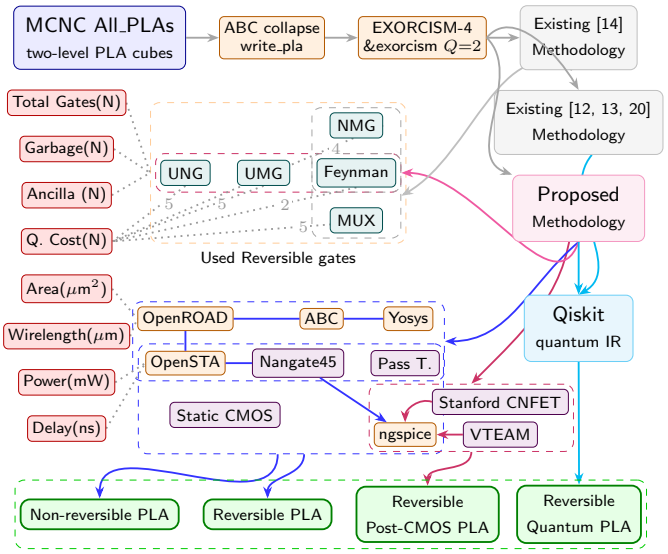


Fig. 7. Experimental workflow. Baselines and Prefix Cube share one cover P; Nangate45, Stanford CNTFET/VTEAM, and Qiskit interpret the proposed netlist $\mathcal{N}$. The Qiskit branch reports gate count, depth, and qubit count.

`.pla` occupancy list [6], [31], [32]. Existing [14], Prefix Cube, NCT [12], [18], the Gaur CNOT/Toffoli factors [13], and the Zhou swap-versus-switch cost [20] all receive this same P. Prefix Cube builds $T(P,\sigma)$ (Algorithms 1–2) and reuses the XOR plane of [8]; no stage rewrites P.

The implementation branches interpret rather than resynthesize $\mathcal{N}$. OpenROAD/OpenSTA/ngspice report area, wirelength, power, and delay under Nangate45 PTL and Stanford CNTFET/VTEAM; Qiskit reads $\mathcal{N}$ as quantum IR. The four realization classes are classical AND–OR, reversible PTL, reversible post-CMOS, and the Qiskit embedding of (3). Power falls only through fewer macros, not cancellation of CV^2 (8).

The 20 representative MCNC PLA benchmarks [6] include `misex1`, the GA/QC extrema, and the largest n/p covers. Covers with overlapping SOP cubes remain fixed occupancy lists. We test all 2^n minterms for $n \le 12$ and 4096 random vectors otherwise; every product appears in the Prefix Cube trie. Scores are resource counters on P, not SAT against a BLIF. Existing mappers are companion Python implementations. Mean reduction is

$$\overline{\Delta}(X) = \frac{1}{|I_E|}\sum_{i\in I_E}\frac{X_i^E - X_i^\dagger}{X_i^E}, \quad I_E = \{i : X_i^E > 0\}. \tag{13}$$

$$\begin{aligned} \mathrm{SW} &= n_{\mathrm{primed}} + n_{\mathrm{SwapLiterals}},\\ \mathrm{CP} &= n_\nabla + n_\triangle + n_{\mathrm{F2G}},\\ \mathrm{RX} &= n_{\mathrm{DOT}}^{\mathrm{AND}} + n_{\mathrm{DOT}}^{\mathrm{XOR}},\\ \mathrm{OP} &= n, \qquad \mathrm{BU} = n_{\mathrm{AND}}. \end{aligned} \tag{14}$$

Prefix Cube executes no SwapLiterals; OP=n preserves x in (3). Medians represent the right-skewed data.

B. Gate, Garbage, Ancilla, and Quantum Cost Metrics

Table VI is the complexity of each mapper on a fixed cover P, written only as closed expressions. Prefix Cube attains $C_\wedge = |V_{\ge 2}|$ in $O(n^2p + pk + pm)$ time and $O(|V| + m)$ memory, with $|V| \le 1 + np$ and no 2^n table (Proposition 1, Theorem 3). Figure 8 reports the measured counts on the 20 circuits: mean GA/AI/QC fall by 53.4/62.3/53.2% versus [14], and mean GB falls by 62.9%. Versus [14], QC is higher only on `rd53`. The outputs are $p = |P|$, $k = \max|c|$, $n = |X|$, m. The product-list row is the product-pair scan of that mapper. Prefix Cube memory uses $|V| \le 1+np$ and does not allocate 2^n (Theorem 3).

`dk17` is the peak GA cut (83.7%), `ryy6` the peak QC cut (94.3%), and `rd53` the QC exception versus [14]. Table VII holds P and the UMG/UNG library fixed.

Table VII compares three mappers on one fixed P. NCT is highest on GA and QC. Prefix Cube is lowest on GA, AI, and QC.

Median runtime is 9.4 ms (maximum 118 ms on `apex3`). Measured times track n^2p (Theorem 3). The mapper never allocates a 2^n table.

Fig. 8 reports that comparison. Mean GA/AI/QC follow $|V_{\ge 2}|$ from Table VI. Mean GB falls by 62.9% versus [14]. Every per-circuit GA point versus [14] lies below $y{=}x$. QC is higher than [14] only on `rd53`.

C. Hardware Metrics and Cost Optimization in Post-CMOS

Stanford 32 nm VS-CNFET and VTEAM 1T1R interpret the same $\mathcal{N}$ (Theorem 4). Milliwatts fall only because fewer UMG/UNG cells are instantiated under (8). On `misex1`, (9)–(10) give $N_{\mathrm{T}}(F) = 603$ MOS transistors and $N_{\mathrm{LRS}}(F) = 77$ programmed 1R cells.

Four PLA classes—non-reversible AND–OR, reversible PTL (Nangate45), reversible post-CMOS (CNFET$^\xi$ of Table V), and the Qiskit embedding—are scored on the same 20 representative names. The PTL and post-CMOS columns are conventional switching of the Prefix Cube netlist, not energy-recovery measurements. Non-reversible PLA uses [5]

$$\begin{aligned} G_{\mathrm{NR}} &= \max(8, \lceil 0.38\,\mathrm{GA}^{\S} + 0.22\,\mathrm{GA}^{\dagger}\rceil),\\ P_{\mathrm{NR}} &= G_{\mathrm{NR}}(0.82{\times}10^{-4}) + \mathrm{AI}^{\dagger}(0.58{\times}10^{-4})\ \mathrm{W}, \end{aligned}$$

where $\mathrm{GA}^{\S}$ sizes the non-reversible gate estimate and $(\mathrm{GA}^{\dagger}, \mathrm{AI}^{\dagger}, \mathrm{QC}^{\dagger})$ are the Prefix Cube counters on the same 20 circuits. The AND/Feynman mix of that netlist is recovered from the 5+1 quantum-cost library,

$$n_{\mathrm{AND}} = \frac{\mathrm{QC}^{\dagger} - \mathrm{GA}^{\dagger}}{4}, \qquad n_{\mathrm{FG}} = \mathrm{GA}^{\dagger} - n_{\mathrm{AND}}. \tag{15}$$

TABLE VI
COMPLEXITY ON A FIXED ESOP COVER P

Mapper	$C_\wedge$, QC	Time	Memory
Existing [14]	product-list AND count	$O(p^2k)$	$O(pk)$
NCT [12], [18]	$n_{\mathrm{CX}} + 5n_{\mathrm{CCX}}$	$\Theta(pk)$	$O(pk)$
Prefix Cube (Proposed)	$\lvert V_{\ge 2}\rvert$, (5)	$O(n^2p{+}pk{+}pm)$	$O(\lvert V\rvert{+}m)$

TABLE VII
THREE MAPPERS ON IDENTICAL FIXED P

Mapper	GA	GB	AI	QC
Existing [14]	2889.6	2055.2	2040.5	9234.3
Existing NCT [12], [18]	3118.8	2731.3	2703.9	13929.0
Prefix Cube (Proposed)	1332.5	736.1	732.4	4228.6

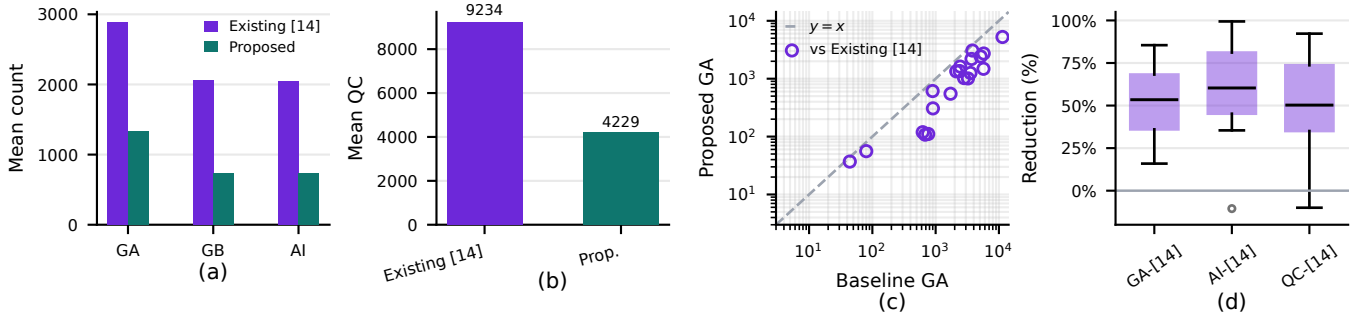


Fig. 8. Prefix Cube versus Existing [14] on 20 representative MCNC PLA circuits. (a) Mean GB, GA, AI. (b) Mean QC. (c) Per-circuit GA on log–log axes. The line $y{=}x$ is a tie. (d) Boxplots of per-circuit reductions (13).

Reversible PTL and post-CMOS switching then scale the Table V macros,

$$P_{\mathrm{PLA}} = n_{\mathrm{AND}}\bar{P}^{\tau}_{\mathrm{AND}} + n_{\mathrm{FG}}P^{\tau}_{\mathrm{FG}},$$
$$P_{\mathrm{post}} = n_{\mathrm{AND}}\bar{P}^{\xi}_{\mathrm{AND}} + n_{\mathrm{FG}}P^{\xi}_{\mathrm{FG}}, \tag{16}$$

with $\bar{P}_{\mathrm{AND}}$ the mean of UMG and UNG on that fabric. The Qiskit column uses a QC-proportional milliwatt proxy $P_{\mathrm{Q}} = \kappa\,\mathrm{QC}^{\dagger}\bar{P}^{\tau}_{\mathrm{AND}}$ (κ=0.32), separate from G, D, and Q. Table XII reports mean PTL / CNFET / proxy powers of 212.85/1.295/275.43 mW. Milliwatts remain conventional CV^2 (8).

D. Qiskit Cost Realization

Qiskit [1] interprets the same $\mathcal{N}$ as a unitary embedding of (3) by Algorithm 3. The quantum-circuit realization is scored before transpilation, by garbage GB, gate count GA, ancilla AI, and macro quantum cost QC (Table IX). NCT [12], [18] is the ESOP Toffoli cascade, with $\mathrm{qc(T)} = 5$ and $\mathrm{qc(CNOT)} = 1$. Gaur and Thapliyal [13] generalize QMA3 onto that cascade: Table I of that paper keeps the CNOT and Toffoli counts and removes $n{+}1$ ancilla lines by zero reset, so GA and QC stay with NCT while GB and AI each fall by $n{+}1$. The modulo adder is not rebuilt, and NMED is not measured. Zhou *et al.* [20] generalize Algorithm 1 onto the same cascade. For each interaction the cheaper of the SWAP path, one switch, and one gate teleportation is kept. On these covers that choice is a switch except for a few SWAP edges. Each selected interaction is one extra gate of $\mathrm{qc} = 1$ and allocates no rail, so GB and AI stay at the NCT values while GA and QC rise by the interaction count. Prefix Cube is the UMG/UNG netlist. On `misex1`, NCT GB/GA/AI/QC are 66/91/57/319, Gaur 57/91/48/319, Zhou 66/181/57/409, and Prefix Cube 20/45/19/113. Table VIII writes the expressions, where N_{link} is the number of interactions for which Algorithm 1 of [20] selects a switch, a SWAP, or a teleportation. Each link has $\mathrm{qc} = 1$ and adds no rail. QMA3 of [13] is the zero-reset rule on the same CNOT/Toffoli cascade; the modulo adder is not rebuilt. Per-circuit values: Table IX.

Fig. 9 reports the NCT comparison on the same covers: mean GB/GA/AI/QC fall by 71.6/60.6/71.3/69.0%, and GA/QC fall on 20/20 names.

Garbage falls because NCT does not reuse prefixes. Peak GA/QC cuts are 85.2% (`apla`) and 94.4% (`ryy6`).

TABLE VIII
QUANTUM REALIZATION EXPRESSIONS ON A FIXED COVER P

Score	Expression
NCT [12], [18]	GB, GA, AI; $\mathrm{QC} = n_{\mathrm{CX}} + 5n_{\mathrm{CCX}}$
Gaur QMA3 [13]	GB and AI minus $n{+}1$; GA and QC as NCT
Zhou Algorithm 1 [20]	GB and AI as NCT; GA and QC plus N_{link}
Prefix Cube QC	(5)

Table IX shows the quantum realization before transpilation. NCT [12] is the ESOP Toffoli cascade. Gaur [13] applies QMA3: CNOT and Toffoli counts stay, and $n{+}1$ zero resets remove that many lines from GB and AI. Zhou [20] applies Algorithm 1 on the logical circuit: each selected link has $\mathrm{qc} = 1$ and adds no rail. Prefix Cube is UMG/UNG. $\mathrm{qc(T)} = \mathrm{qc(UMG)} = \mathrm{qc(UNG)} = 5$ and $\mathrm{qc(CNOT)} = 1$. No Prefix Cube row raises GA, GB, or QC versus NCT. Expressions: Table VIII. Transpiled G, D, Q: Table X.

Fig. 10(a) draws that quantum cost, with covers ordered by increasing NCT QC.

Table IX is that realization. Mean Gaur GB/AI are 2707.0/2679.7, against NCT 2731.3/2703.9, because each circuit drops $n{+}1$ zero-reset lines. Mean Zhou GA/QC are 6232.3/17042.5, against NCT 3118.8/13929.0, because each selected link is one extra gate. No Prefix Cube row raises GA, GB, or QC versus NCT. A published quantum PAL [33] drops from QC 53 to 28 when remapped as Prefix Cube AND+XOR.

Gaur markers lie on the NCT line because QMA3 keeps the CNOT and Toffoli counts. Zhou lies above that line. Prefix Cube lies below [12], [13], and [20] on every cover. Panel (b)

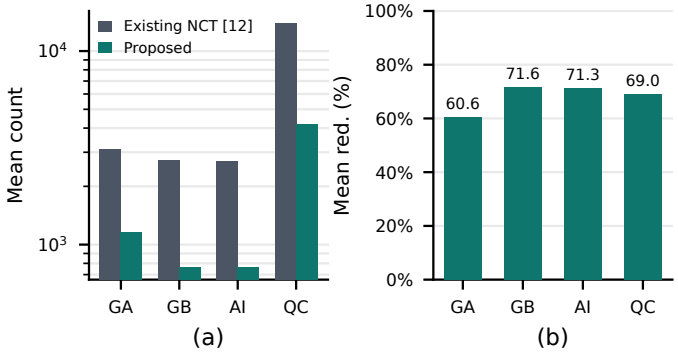


Fig. 9. Existing NCT [12] versus Prefix Cube on 20 representative MCNC PLA circuits. (a) Mean GA, GB, AI, QC. (b) Mean reduction (13). Cost expressions: Table VIII.

TABLE IX
QUANTUM REALIZATION: NCT [12], GAUR [13], AND ZHOU [20] VERSUS PREFIX CUBE

Name	NCT [12]				Gaur [13]				Zhou [20]				Prefix Cube			
	GB	GA	AI	QC	GB	GA	AI	QC	GB	GA	AI	QC	GB	GA	AI	QC
alu4	4440	5084	4420	22764	4425	5084	4405	22764	4440	10162	4420	27842	1777	2421	1771	9501
apex1	3219	4110	3174	16758	3173	4110	3128	16758	3219	8208	3174	20856	1038	1929	1036	5969
apex3	3617	4208	3559	18432	3562	4208	3504	18432	3617	8409	3559	22633	1181	1772	1177	6444
apla	676	751	668	3423	665	751	657	3423	676	1504	668	4176	36	111	38	263
b2	2305	3016	2290	12172	2288	3016	2273	12172	2305	6031	2290	15187	782	1493	783	4585
b3	2605	2912	2561	13144	2572	2912	2528	13144	2605	5810	2561	16042	709	1017	697	3781
chkn	1610	1734	1559	7970	1580	1734	1529	7970	1610	3447	1559	9683	425	550	404	2166
cordic	17077	18234	17033	86366	17053	18234	17009	86366	17077	36447	17033	104579	3625	4782	3604	19198
dk17	335	386	326	1678	324	386	315	1678	335	770	326	2062	61	112	62	344
duke2	966	1145	951	4933	943	1145	928	4933	966	2293	951	6081	442	621	449	2385
e64	2145	2145	2080	10465	2079	2145	2014	10465	2145	4291	2080	12611	1366	1367	1367	6835
f51m	1800	2816	1792	9984	1791	2816	1783	9984	1800	5631	1792	12799	9	1025	9	1041
ibm	816	941	737	3889	767	941	688	3889	816	1851	737	4799	182	307	151	911
misex1	66	91	57	319	57	91	48	319	66	181	57	409	20	45	19	113
misex3c	1505	1746	1491	7710	1490	1746	1476	7710	1505	3493	1491	9457	637	879	638	3395
pdc	3744	4178	3752	19182	3727	4178	3735	19182	3744	8379	3752	23383	459	893	483	2809
rd53	128	158	121	642	122	158	115	642	128	314	121	798	49	79	47	267
ryy6	543	639	512	2687	526	639	495	2687	543	1262	512	3310	23	119	8	151
table3	3651	4171	3637	18719	3636	4171	3622	18719	3651	8341	3637	22889	1538	2058	1538	8186
table5	3378	3912	3359	17344	3360	3912	3341	17344	3378	7821	3359	21253	966	1500	964	5328
Mean	2731.3	3118.8	2703.9	13929.0	2707.0	3118.8	2679.7	13929.0	2731.3	6232.3	2703.9	17042.5	766.2	1154.0	762.2	4183.6

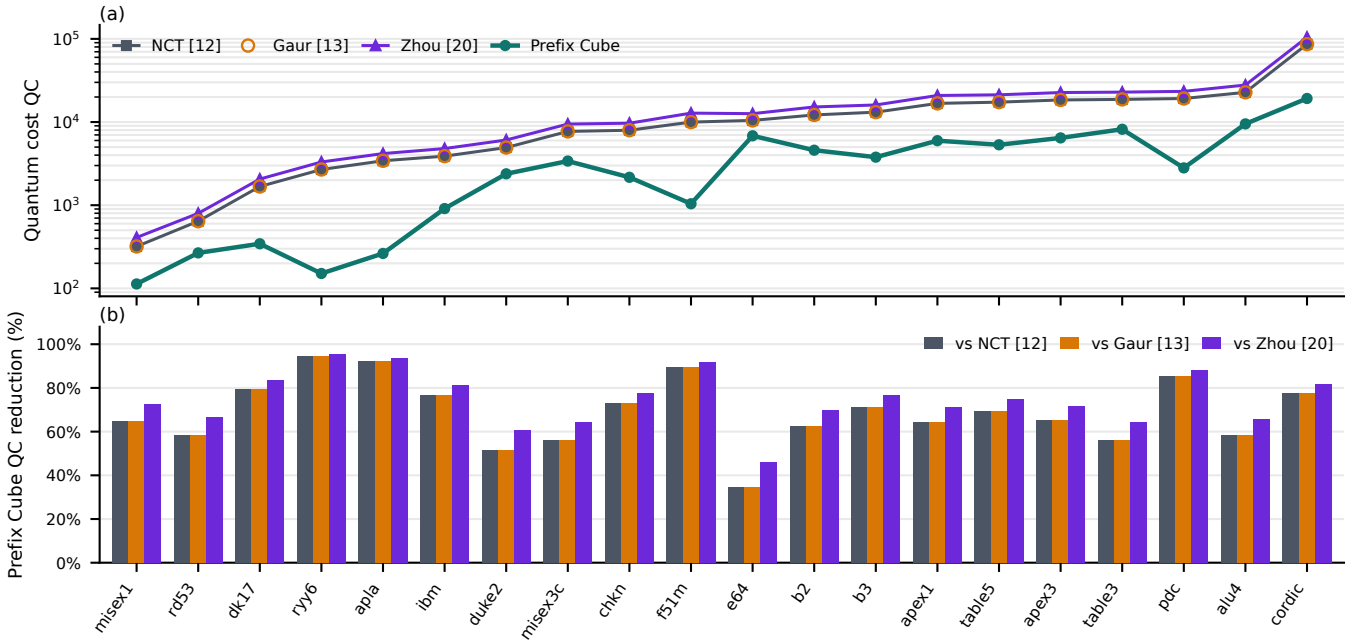


Fig. 10. Quantum cost of Table IX, ordered by increasing NCT QC. (a) NCT [12], Gaur [13], Zhou [20], and Prefix Cube. Gaur uses the same QC as NCT. (b) Vertical bars of the Prefix Cube QC reduction on the same covers. The Gaur bars match NCT because QMA3 keeps the CNOT and Toffoli counts.

shows the same comparison as vertical bars of the Prefix Cube QC reduction (13): the mean is 69.0% versus NCT and Gaur, and 74.8% versus Zhou.

After the realization, qiskit-transpile scores gate count G, depth D, and qubit count Q (`optimization_level=2`, `seed_transpiler=42`, basis $\{x, cx, ccx, swap\}$). Table X uses only the NCT cascade of [12]. NCT Q is the number of qubits of the circuit built from that benchmark. Its Status column is the circuit type after the size test: *full* when both transpiles run, and *logical* when both are skipped. The Impact columns are the change from that logical circuit to the reported NCT row. They are zero on a logical row. On a full row the pass leaves Q fixed and cancels polarity-X gates only. Prefix Cube G is the UMG/UNG/CX count of $\mathcal{N}$ (1154.0 mean, versus NCT reversible GA 3118.8). Prefix Cube Q is the Bennett width $n+\mathrm{AI}$. Prefix Cube AND-plane depth is at most $n-1$. Figure 6 is `misex1`: NCT G=134, D=46, Q=38 versus Prefix Cube G=45, $D_\wedge \leq 7$, Q=27. Table XI is the CX/CCX tally (mean 3113.6) and the macro cost $n_{\mathrm{CX}} + 5n_{\mathrm{CCX}}$ (mean 13929.0). On `misex1`, Shende $n_{\mathrm{CX}} + n_{\mathrm{CCX}} = 90$ against Prefix Cube QC 113. Table X is the qiskit-transpile ledger, not the library quantum cost. Baseline: [12] only. NCT Q is the qubit count of the benchmark circuit; qiskit-transpile does not change it (ΔQ=0). ΔG and

ΔD are the NCT gate-count and depth change. *full* means both calls ran, NCT on $\{x, cx, ccx, swap\}$ and Prefix Cube on $\{x, cx, umg, ung, swap\}$ (`dk17`, `misex1`, `rd53`). Their logical NCT (G, D) values are $912/118$, $188/58$, and $292/93$; CX and CCX stay, and only polarity X is removed. The UMG/UNG call has the same ΔG and ΔD. *logical* means both calls were skipped. Prefix Cube G is the UMG/UNG/CX count on every row; $D_\wedge = n - 1$; $Q = n +$ AI. Mean G falls by 82.2%. Library GB, GA, AI, and QC: Table IX.

VI. System-Level Embedding

The proposed live deployment is an energy-harvest-assisted wearable emergency monitor, not an autonomous diagnostic device. Figure 11(a) is the Qiskit realization of its medical decision core (Algorithm 3): quality-gated basis flags h, o, f, i and validity flags $q_{\mathrm{ECG}}, q_{\mathrm{PPG}}$ enter Prefix Cube UMG/UNG gates; Feynman CNOTs copy cardiac, hypoxia, fall, and combined-risk products p_c, p_h, p_f, and p_m to persistent event rails; and $U_{\mathrm{AND}}^{\dagger}$ returns working ancilla to $|0\rangle$ before computational-basis event measurement. Figure 11(b) exposes that circuit as four reusable blocks: (A) basis/quality encoding, (B) Prefix Cube products, (C) Feynman copy plus uncomputation, and (D) event readout.

Figure 12 places the same A–D blocks in the wearable pathway. Multimodal ECG, PPG/SpO_2, temperature, and inertial sensors pass an analog front-end quality gate into the always-on reversible core. A duty-cycled TinyML verifier checks temporal context and uncertainty; a confirmed event preserves a signal window and escalates along local, caregiver, or emergency tiers. An energy-autonomous supply, a signed clinical profile of thresholds on the fixed ESOP cover P, and a trust supervisor (secure boot, CRC, BIST, watchdog) sit above the datapath. The quantum path is a shadow lane of the same netlist $\mathcal{N} \mapsto U_{\mathrm{RPLA}}$, physically outside the fail-safe classical loop, for hybrid or offline evaluation of the medical core. Cryogenic control, decoder latency, and surface-code qubits are not included [21], [22]; system cost of the core remains the (G, D, Q) triple of Table X. The monitor assists potentially life-saving early warning; it does not autonomously diagnose or treat.

Those products are why Prefix Cube sharing is the live CAD fact, not a later rewrite of the island. Cardiac $p_c = hq_{\mathrm{ECG}}$ and hypoxia $p_h = oq_{\mathrm{PPG}}$ each take one UMG that already folds the quality flag, so a poor lead cannot raise an event; combined-risk $p_m = p_c o$ taps p_c instead of rebuilding hq_{ECG}, which is (1) on this cover. The reversible island is always-on while TinyML is duty-cycled, so harvested thermoelectric/kinetic milliwatts track n_{AND}: the same mapper cuts mean GA/AI by $53.4/62.3\%$ versus the product-list RPLA of [14], and the CNFET reading of one netlist is $1.295\,\mathrm{mW}$ mean switching against $212.85\,\mathrm{mW}$ PTL (Table XII), still conventional CV^2 from fewer macros. AND-plane depth stays $D_\wedge \le n - 1$ (mean 22.3 versus NCT depth 907.5), mean critical-path delay on the 20 covers is 0.33–0.49 ns, and a signed-profile retarget of P takes median $9.4\,\mathrm{ms}$ of host CAD without a 2^n table, so personalization and rollback remain in the trust plane of Fig. 12.

TABLE X
Qiskit Transpile Cost: NCT [12] Versus Prefix Cube

Name	NCT [12]			Prefix Cube			Impact		
	G	D	Q	G	$D_\wedge$	Q	ΔG	ΔD	Status
`alu4`	10344	1974	685	2421	13	1785	0	0	logical
`apex1`	7500	1373	568	1929	44	1081	0	0	logical
`apex3`	8735	1710	692	1772	53	1231	0	0	logical
`apla`	1913	251	112	111	9	48	0	0	logical
`b2`	5960	1050	351	1493	15	799	0	0	logical
`b3`	6850	1001	361	1017	31	729	0	0	logical
`chkn`	3479	451	206	550	28	433	0	0	logical
`cordic`	37395	3517	1220	4782	22	3627	0	0	logical
`dk17`	482	87	90	112	9	72	−430	−31	full
`duke2`	2508	420	185	621	21	471	0	0	logical
`e64`	6178	524	258	1367	64	1432	0	0	logical
`f51m`	4864	771	278	1025	7	17	0	0	logical
`ibm`	1586	476	246	307	47	199	0	0	logical
`misex1`	134	46	38	45	7	27	−54	−12	full
`misex3c`	3529	817	336	879	13	652	0	0	logical
`pdc`	9679	1058	418	893	15	499	0	0	logical
`rd53`	230	73	46	79	4	52	−62	−20	full
`ryy6`	1872	342	133	119	15	24	0	0	logical
`table3`	8277	1148	407	2058	13	1552	0	0	logical
`table5`	8329	1061	361	1500	16	981	0	0	logical
Mean	6492.2	907.5	349.6	1154.0	22.3	785.6	−27.3	−3.15	—

TABLE XI
Qiskit Transpile Costs: NCT [12] Versus Prefix Cube

Realization	Formula	Mean	$\overline{\Delta}$
NCT [12]	$n_{\mathrm{CX}} + n_{\mathrm{CCX}}$	3113.6	—
NCT macro [12], [18]	$n_{\mathrm{CX}} + 5n_{\mathrm{CCX}}$	13929.0	69.0%
Prefix Cube	(5)	4183.6	—

$\overline{\Delta}$ is (13) of Prefix Cube macro QC against the NCT macro row. The tally $n_{\mathrm{CX}} + n_{\mathrm{CCX}}$ is not charged at $\mathrm{qc} = 5$. Primary G, D, Q: Table X.

VII. Conclusion

Prefix Cube treats prefix sharing as a structural resource for reversible programmable logic array (RPLA) optimization: instead of reconstructing product-term prefixes during gate mapping, it transforms a fixed Exclusive-OR sum-of-products (ESOP) cover P into a greedily ordered prefix-sharing tree before any device map. Shared-prefix reuse is then explicit and measurable. The AND-plane cost is $C_\wedge = |V_{\ge 2}|$ with tree-size bound $|V| \le 1 + np$ (1). Algorithms 1–3 with ConstructXORPlane run in $O(n^2 p + pk + pm)$ time, so the construction scales with the cover parameters rather than a 2^n enumeration. Product-list sorts and NCT Toffoli cascades rebuild prefixes that the tree shares once.

On 20 representative MCNC PLA circuits, mean GA, GB, AI, and QC fall by 53.4%, 62.9%, 62.3%, and 53.2% versus [14]; mean GA/AI/QC fall by $60.6/71.3/69.0\%$ versus the NCT cascade of [12], [18]. Versus the Qiskit NCT flow, mean gate count falls from 6492.2 to 1154.0 (82.2%) and AND-plane depth is bounded by $n - 1$ (mean 22.3 versus NCT depth 907.5, a 97.5% cut). Shared prefixes occupy dedicated rails, so mean qubit count is 785.6 versus 349.6 (124.7% more qubits). Those figures mark a frontier among gate economy, logical depth, and qubit resources rather than a single-metric minimum.

Algorithm 3 obtains Bennett outputs by computational-basis measurement after uncomputation. Surface-code overhead is outside this CAD, and no quantum speedup is claimed [21],

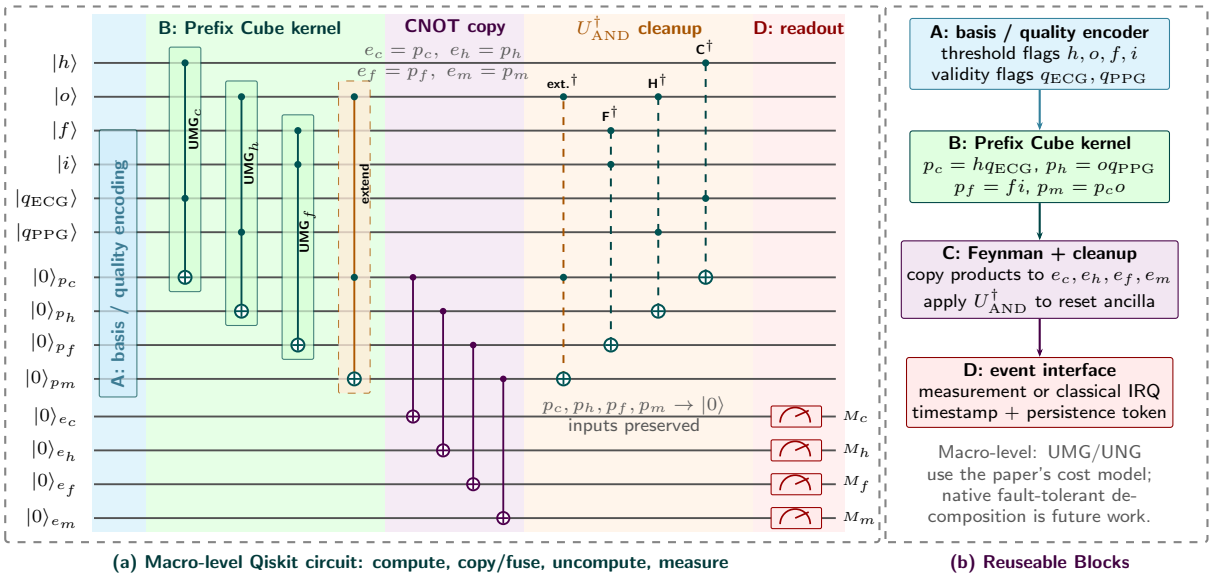


Fig. 11. Qiskit realization of the Prefix Cube medical decision core (Algorithm 3). (a) UMG/UNG macros compute shared cardiac, hypoxia, fall, and combined-risk products; Feynman CNOTs copy them to event rails; and $U^\dagger_{\mathrm{AND}}$ resets prefix ancilla before computational-basis measurement. (b) Blocks A–D are reused in the wearable pathway of Fig. 12. Surface-code decomposition is not included.

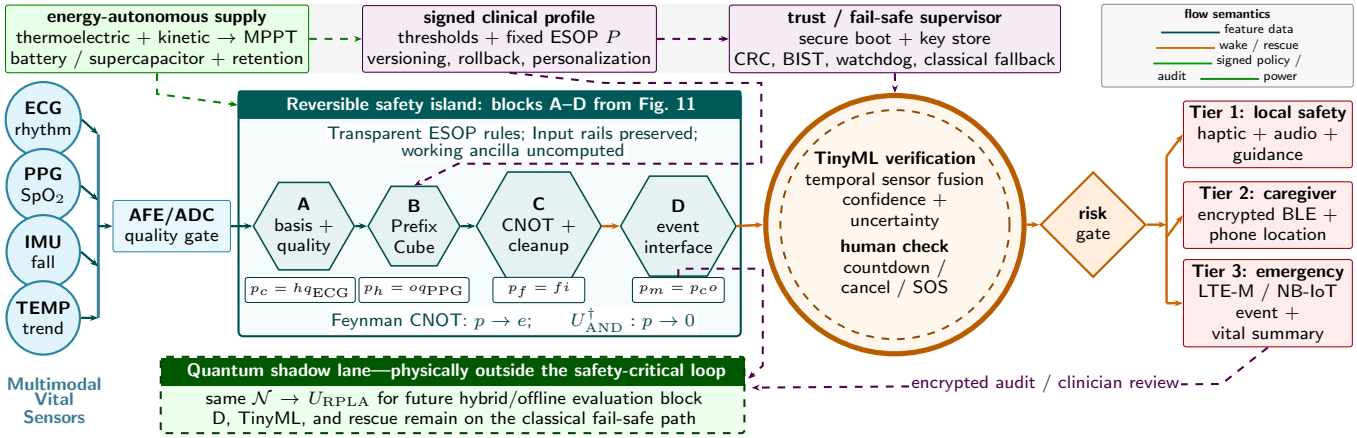


Fig. 12. Live wearable emergency-monitoring pathway using blocks A–D of Fig. 11(b). Quality-gated ECG, PPG/SpO_2, temperature, and inertial sensors drive the always-on reversible core; TinyML verifies temporal context before graded local, caregiver, or emergency alerts. The quantum shadow lane evaluates the same netlist $\mathcal{N}$ outside the classical fail-safe loop. The system assists early warning and does not autonomously diagnose or treat.

[22]. Those bounds separate the demonstrated synthesis and resource cuts from fault-tolerant hardware performance that this study does not establish.

One netlist $\mathcal{N}$ is interpreted as Nangate45 PTL, Stanford CNTFET/VTEAM, and a Qiskit cost model. Table XII reports mean switching power of 155.22, 212.85, 1.295, and 275.43 mW for the non-reversible PLA, reversible PTL, CNTFET, and the Qiskit proxy, with mean critical-path delay 0.37, 0.39, 0.49, and 0.33 ns. Structural reversible-logic cuts do not imply uniform power or delay across fabrics; the PTL and CNTFET milliwatts remain conventional CV^2 estimates, not quantum-hardware energy.

Prefix Cube therefore reduces redundant AND-plane construction at the representation level, before physical or quantum-device costs are introduced. The contribution is a bounded, explicit route from a fixed ESOP cover to a shared netlist whose logical, resource, and technology-dependent properties can be scored independently. That separation is a reproducible basis for adaptive literal ordering, multi-objective gate–depth–qubit optimization, and technology-aware reversible synthesis for quantum and post-CMOS computing.

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TABLE XII
ESTIMATED CONVENTIONAL SWITCHING POWER AND CRITICAL-PATH DELAY ON 20 REPRESENTATIVE MCNC PLA CIRCUITS: NON-REVERSIBLE PLA, REVERSIBLE PTL ($GA^{\dagger}$), REVERSIBLE POST-CMOS ($CNFET^{\xi}$, TABLE V), AND QISKIT EMBEDDING ($QC^{\dagger}$ PROXY). PTL AND POST-CMOS MILLIWATTS ARE LIBRARY CV^2 ESTIMATES (8), NOT ADIABATIC OR LANDAUER-LIMITED ENERGY.

Name	Switching Power (mW)				Critical-Path Delay (ns)			
	P_{NR}	P_{PLA}	P_{post}	P_{Q}	$\mathrm{CP}_{\mathrm{NR}}$	$\mathrm{CP}_{\mathrm{PLA}}$	$\mathrm{CP}_{\mathrm{post}}$	CP_{Q}
alu4	305.04	430.22	2.579	618.79	0.53	0.59	0.73	0.51
apex1	205.10	325.31	2.001	385.21	0.46	0.56	0.70	0.43
apex3	221.60	408.58	2.556	413.35	0.49	0.65	0.81	0.44
apla	23.360	15.590	0.097	17.131	0.20	0.15	0.19	0.14
b2	244.75	233.74	1.425	298.00	0.53	0.47	0.59	0.39
b3	157.99	175.09	1.053	245.89	0.42	0.40	0.50	0.37
chkn	86.490	97.944	0.587	141.08	0.32	0.31	0.38	0.30
cordic	665.12	921.82	5.536	1309.56	0.75	0.82	1.02	0.67
dk17	25.914	17.186	0.105	22.146	0.20	0.15	0.19	0.15
duke2	58.550	97.227	0.592	125.06	0.25	0.32	0.40	0.29
e64	165.50	270.07	1.594	429.12	0.37	0.45	0.56	0.45
f51m	106.71	110.67	0.720	67.807	0.40	0.40	0.50	0.23
ibm	42.132	47.522	0.290	59.339	0.24	0.24	0.29	0.22
misex1	4.488	7.273	0.046	7.035	0.09	0.11	0.14	0.10
misex3c	318.56	442.12	2.683	581.86	0.56	0.62	0.77	0.50
pdc	158.70	178.89	1.107	200.81	0.44	0.44	0.54	0.34
rd53	3.186	5.996	0.036	7.881	0.07	0.09	0.12	0.11
ryy6	22.030	13.572	0.087	9.836	0.20	0.16	0.19	0.12
table3	153.35	252.58	1.542	318.52	0.40	0.49	0.61	0.40
table5	135.80	205.60	1.260	250.19	0.39	0.45	0.56	0.37
Mean	155.22	212.85	1.295	275.43	0.37	0.39	0.49	0.33

DECLARATION OF DATA AVAILABILITY

MCNC benchmark circuits used in this paper are available at https://github.com/ispras/hdl-benchmarks. The benchmark analysis experiments, including the Prefix Cube mapping and the Qiskit cost comparisons on those covers, are available at https://github.com/sajibmitra/benchmark-pla.

APPENDIX A
EXPERIMENTAL ENVIRONMENT

Experiments used the open-source logic-synthesis, place-and-route, circuit-simulation, and (optionally) quantum-circuit toolchain below.

a) Host.: Experiments ran on Ubuntu 24.04.4 LTS (kernel 6.8), aarch64 (ARM Neoverse-N1), with 2 CPU cores and 3.7 GiB RAM, using Python 3.12.3.

b) EDA toolchain.: The host-native toolchain is summarized in Table XIII.

c) Technology library.: PTL and TGL metrics use Nangate45 ($V_{\mathrm{DD}} = 1.8\,\mathrm{V}$) from OpenROAD: area ($\mu\mathrm{m}^2$), average power ($\times 10^{-4}\,\mathrm{W}$), and $t_{\mathrm{PLH}}/t_{\mathrm{PHL}}$ (ps). Post-CMOS results use VTEAM and Stanford CNFET models in ngspice.

d) Flow.: SOP/ESOP netlists are mapped with Yosys (Nangate45), OpenROAD, OpenSTA, and ngspice. Qiskit interprets the same netlist as a circuit scored by gate count, depth, and qubit count.

REFERENCES

[1] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood, J. Lishman, J. Gacon, S. Martiel, P. D. Nation, L. S. Bishop, A. W. Cross, B. R. Johnson, and J. M. Gambetta, "Quantum computing with Qiskit," *arXiv:2405.08810*, 2024.

[2] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM J. of R&D*, vol. 5, no. 3, pp. 183–191, 1961.

[3] C. Bennett, "Logical reversibility of computation," *IBM J. of Research and Development*, vol. 17, no. 6, pp. 525–532, 1973.

[4] H. Fleisher and L. Maissel, "An introduction to array logic," *IBM J. of Res. and Development*, vol. 19, no. 2, pp. 98–109, 1975.

[5] J. M. Rabaey, A. Chandrakasan, and B. Nikolic, *Digital Integrated Circuits: A Design Perspective*, 2nd ed. Prentice Hall, 2003.

[6] S. Yang, "Logic synthesis and optimization benchmarks user guide, version 3.0," Microelectronics Center of North Carolina (MCNC), Tech. Rep., 1991, mCNC/EPFL benchmark circuits used under EDA_BENCHMARKS_ROOT.

[7] R. Feynman, "Quantum mechanical computers," *Foundations of Physics*, vol. 16, no. 6, pp. 507–531, 1986.

[8] S. K. Mitra and M. R. Rahman, "Reversible programmable logic arrays," in *Proceedings of the 2016 29th Int. Conf. on VLSI Design and 2016 15th Int. Conf. on Embedded Systems (VLSID)*. New York, NY, USA: ACM, 2016, pp. 379–384. [Online]. Available: https://dl.acm.org/doi/abs/10.5555/2924725.2925251

[9] A. Barenco, C. H. Bennett, R. Cleve, D. P. DiVincenzo, N. Margolus, P. Shor, T. Sleator, J. A. Smolin, and H. Weinfurter, "Elementary gates for quantum computation," *Physical Review A*, vol. 52, no. 5, pp. 3457–3467, 1995.

[10] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. Cambridge, U.K.: Cambridge University Press, 2010.

[11] H. M. Hasan Babu, *Quantum Computing: A Pathway to Quantum Logic Design*, 2nd ed. Bristol, U.K.: IOP Publishing, 2020.

[12] V. V. Shende, S. S. Bullock, and I. L. Markov, "Synthesis of quantum-logic circuits," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 25, no. 6, pp. 1000–1010, 2006.

[13] B. Gaur and H. Thapliyal, "Novel optimized designs of modulo 2n+1 adder for quantum computing," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 32, no. 9, pp. 1759–1763, 2024.

[14] N. Tara and H. M. H. Babu, "Synthesis of reversible pla using products sharing," *Journal of Computational Electronics*, vol. 15, no. 2, pp. 420–428, 2016. [Online]. Available: https://doi.org/10.1007/s10825-015-0762-5

[15] D. M. Miller, D. Maslov, and G. W. Dueck, "A transformation based algorithm for reversible logic synthesis," *Proc. Design Automation Conf. (DAC)*, pp. 318–323, 2003.

[16] M. Soeken and et al., "Hierarchical reversible logic synthesis using LUTs," in *Proc. Design Automation Conf. (DAC)*, 2017.

[17] M. Soeken, R. Wille, G. W. Dueck, and R. Drechsler, "From reversible logic to quantum circuits: Logic design for an emerging technology," in *Proc. Int. Conf. Embedded Computer Systems: Architectures, Modeling and Simulation (SAMOS)*, 2016, pp. 268–274.

[18] K. Fazel, M. A. Thornton, and J. E. Rice, "ESOP-based Toffoli gate cascade generation," in *Proc. IEEE Pacific Rim Conf. Communications, Computers and Signal Processing*, 2007, pp. 206–209.

[19] R. Wille, D. Große, L. Teuber, G. W. Dueck, and R. Drechsler, "RevLib: An online resource for reversible functions and reversible circuits," in *Proc. IEEE Int. Symp. Multiple-Valued Logic (ISMVL)*, 2008, pp. 220–225.

[20] Y. Zhou, K. Liu, H. Luo, L. Xiong, Y. Zhu, E. Casey, J. Cheng, S. Yen-Chi Chen, and Z. Liang, "Eddqc: Enhanced dynamical distributing quantum compilation," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 34, no. 1, pp. 93–103, 2026.

[21] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, "Surface codes: Towards practical large-scale quantum computation," *Physical Review A*, vol. 86, no. 3, p. 032324, 2012.

[22] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[23] R. Zhou and et al., "Transistor realization of reversible "ZS" series gates and reversible array multiplier," *Microelectronics Journal*, vol. 42, no. 2, pp. 305–315, 2011.

[24] S. Kvatinsky and et al., "Vteam: A general model for voltage-controlled memristors," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 62, no. 8, pp. 786–790, 2015.

[25] C.-S. Lee and et al., "A compact virtual-source model for carbon nanotube fets in the sub-10-nm regime—part ii: Extrinsic elements, performance assessment, and design optimization," *IEEE Transactions on Electron Devices*, vol. 62, no. 9, pp. 3070–3078, 2015.

[26] D. B. Strukov, G. S. Snider, D. R. Stewart, and R. S. Williams, "The missing memristor found," *Nature*, vol. 453, no. 7191, pp. 80–83, 2008.

[27] T. Toffoli, "Reversible computing," in *Lecture Notes in Comp. Sci. on Auto., Lang. and Prog.*, vol. 85, 1980, pp. 632–644.

TABLE XIII
EDA TOOLCHAIN USED FOR THE EXPERIMENTS.

Tool	URL	Role	Version / build
Yosys	https://github.com/YosysHQ/yosys	Logic synthesis/tech-map	`v0.63-201-gf7b19774b`
ABC	https://github.com/berkeley-abc/abc	Logic optimization	`80c8a9a19` (2026-04-06)
EXORCISM4	https://github.com/lsils/exorcism	convert pla to esop	`af2497e`
OpenROAD	https://github.com/The-OpenROAD-Project/OpenROAD.git	Floorplan/place/global route	`d34d035` (2026-04-07)
OpenSTA	https://github.com/The-OpenROAD-Project/OpenSTA	Static timing	`12bbfed5` (2026-04-07)
ngspice	https://github.com/ngspice/ngspice	Transient simulation	`032b1c3` (2015-03-08)
Qiskit	https://github.com/Qiskit/qiskit	Quantum-circuit interpreter	`2.3.1`
Xschem	https://github.com/StefanSchippers/xschem	Schematic preview	V3.4.8RC

[28] E. Fredkin and T. Toffoli, "Conservative logic," *International Journal Theoretical Physics*, vol. 21, pp. 219–253, 1982.

[29] A. Peres, "Reversible logic and quantum computers," *Physical Review A*, vol. 32, no. 6, pp. 3266–3276, Dec 1985.

[30] A. Sayem and et al., "Fault-tolerant reversible circuits," in *Proceedings of IEEE International Conference on Computer Science and Automation Engineering (CSAE'11)*, Shanghai, China, 2011, pp. 251–255.

[31] R. Brayton and et al., "ABC: An academic industrial-strength verification tool," in *Computer Aided Verification (CAV)*, 2010, pp. 24–40.

[32] A. Mishchenko and et al., "Fast heuristic minimization of exclusive-sums-of-products," in *Proc. 5th Int. Workshop on Applications of the Reed-Muller Expansion in Circuit Design*, 2001, pp. 242–251, eXORCISM-4 heuristic; invoked via Berkeley ABC `&exorcism` in SOP→ESOP conversion.

[33] F. Akter and et al., "A optimized reversible design for programmable array logic based on quantum computing nanotechnology," in *Innovations in Electrical and Electronics Engineering*, A. Kalam, S. Mekhilef, and S. S. Williamson, Eds. Singapore: Springer Nature Singapore, 2025, pp. 355–367.



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